

Differential transcendence of the Hahn–Exton q -cosine and the arithmetic of its zeros

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July 17, 2026

Abstract

Let $G(q, z) = {}_1\phi_1(0; q; q^2, z) = \sum_{k \geq 0} (-1)^k q^{k(k-1)} z^k / (q; q)_{2k}$, the Hahn–Exton (third Jackson) q -cosine. We prove that for every $0 < q < 1$ the difference Galois group over $\mathbb{C}(z)$ of the second-order q -difference equation satisfied by G contains SL_2 , and that the equation admits no $\sigma\delta$ -integrability relation; via the parametrized difference Galois theory of Hardouin–Singer and Dreyfus–Hardouin–Roques it follows that $G(q, \cdot)$ is differentially transcendental over $\mathbb{C}(z)$. We complement this function-level result with an exact evaluation of the regularized product of the positive zeros, $\prod_{k \geq 1} z_k(q) q^{2(k-1)} = (q; q^2)_\infty$, a modular value. We then pose, and give effective evidence for, the arithmetic question that these results frame: the transcendence of the zeros $z_k(q)$ at algebraic q . This question specializes to the transcendence of a specific reflection-group growth constant β_2 (OEIS A396406), for which we prove an unconditional effective non-rationality bound.

1 Introduction

The three Jackson q -analogues of the Bessel function are among the most-studied special functions of q -series. The third, due to Hahn and Exton, has the q -cosine and q -sine specializations

$$G(q, z) := {}_1\phi_1(0; q; q^2, z) = \sum_{k \geq 0} \frac{(-1)^k q^{k(k-1)}}{(q; q)_{2k}} z^k,$$

which is, up to an explicit prefactor, $J_{-1/2}^{(3)}(\sqrt{z}/q; q^2)$; it is entire of order 0 in z , and its positive zeros $0 < z_1(q) < z_2(q) < \dots$ are real and simple [6]. The analytic theory of these zeros — orthogonality, interlacing, asymptotics — is classical. Their *arithmetic* is, to our knowledge, untouched: nothing is known about the transcendence of $z_k(q)$ at algebraic q , the q -analogue of Siegel’s theorem that the zeros of the Bessel function J_ν are transcendental.

This paper establishes the function-theoretic groundwork for that arithmetic and states the resulting transcendence problem precisely. Our results are of three kinds.

Galois theory (§3). We compute enough of the difference Galois structure of the defining equation to conclude differential transcendence.

Theorem 1. *For every $0 < q < 1$, the Hahn–Exton q -cosine $G(q, \cdot)$ satisfies no algebraic differential equation over $\mathbb{C}(z)$.*

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The engine is an unconditional proof that the difference Galois group contains SL_2 (Theorem 5), obtained by ruling out rank-1 submodules of every symmetric power through a growth estimate on asymptotic-expansion coefficients, together with the absence of a $\sigma\delta$ -integrability relation (Proposition 6).

A modular zero product (§4). Writing $Q = q^2$,

Theorem 2. $\prod_{k \geq 1} z_k(q) Q^{k-1} = (q; q^2)_\infty$, and $G(q, z)/\theta_Q(-z) \rightarrow 1/(q; q)_\infty$ along the negative axis, where $\theta_Q(-z) = \sum_n (-1)^n q^{n(n-1)} z^n$.

The regularized product of the zeros is thus an explicit modular value — transcendental at algebraic q by Nesterenko's theorem [7].

Arithmetic of the zeros (§5). These results frame a clean conjecture.

Conjecture 3 (Z). For algebraic $q \in (0, 1)$, every positive zero $z_k(q)$ of $G(q, \cdot)$ is transcendental.

We explain why Conjecture 3 lies just past the reach of the q -analogue of the Siegel–Shidlovskii method (the double-Pochhammer denominator $(q; q)_{2k}$ places it below the Siegel margin of [1]), and we record an unconditional partial result for the specific zero of number-theoretic interest.

The motivating instance is the growth constant $\beta_2 = 1.4916177871\dots$ of the geodesic growth series of a planar right-triangle reflection group (OEIS A396406), which equals $1/\sqrt{q^*}$ where q^* is the least positive root of $S(q) := G(q, 2q(1-q))$; equivalently $z^* := 2q^*(1-q^*) = z_1(q^*)$ is the smallest positive zero of $G(q^*, \cdot)$. Conjecture 3 implies β_2 transcendental. We prove unconditionally (Proposition 20) that q^* has no rational representation of denominator $\leq 10^{191}$.

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2 The q -difference module

Throughout $Q = q^2$, $\sigma f(z) = f(Qz)$, and $a(z) = q + 1 - qz$. From the ${}_1\phi_1$ recurrence, G satisfies

$$q G(z) - a(z) G(Qz) + G(Q^2 z) = 0, \quad L := q - a\sigma + \sigma^2, \quad (1)$$

a second-order q -difference equation with companion matrix $A(z) = \begin{pmatrix} 0 & 1 \\ -q & a \end{pmatrix}$, $\det A = q$, regular (Poincaré-type) at $z = 0$ and irregular at $z = \infty$ (Newton slopes $\{0, 1\}$). A second solution is $\tilde{G}(z) = z^{1/2} H(z)$ with $H(z) = \sum_{k \geq 0} (-1)^k q^{k^2} (1-q) z^k / (q; q)_{2k+1}$; the pair (G, H) realizes the Hahn–Exton q -cosine/ q -sine $J_{\mp 1/2}^{(3)}$. The Casoratian $\mathrm{Cas}(G, \tilde{G})(z) := G(z)\tilde{G}(Qz) - G(Qz)\tilde{G}(z) = c z^{1/2}$ is nonzero, and the exact q -Wronskian identity

$$q G(z) H(Qz) - G(Qz) H(z) = q - 1 \quad (2)$$

holds (verified to 40 digits). We write M for the associated rank-2 σ -module over $\mathbb{C}(z)$.

At $z = \infty$ the equation has a formal solution basis reflecting the two Newton slopes. With $\theta_Q(x) = \sum_{n \in \mathbb{Z}} Q^{n(n-1)/2} x^n$, these are

$$f_1(z) = \theta_Q(-z) \hat{g}(z), \quad \hat{g}(z) = \sum_{k \geq 0} b_k z^{-k}, \quad b_k = \frac{q^{k-k(k-1)/2}}{(q; q)_k}; \quad f_2(z) = \frac{h(z)}{\theta_Q(-z/q)}, \quad h(z) = \sum_{k \geq 0} c_k z^{-k},$$

where $\hat{g}$ is q -Gevrey divergent (its q -Borel transform is $1/(q^2/\xi; q)_\infty$, so f_1 is Borel–Laplace summable off one Stokes direction) and h is convergent ($c_k = O(Q^{k^2/2})$). The coefficients b_k , which drive the Galois argument below, satisfy $b_k \asymp q^{-k^2/4}$; both $\hat{g}$ and h are determined by (1) and $b_0 = c_0 = 1$. Full details of the resummation are in [3]; here only the asymptotic expansions are used.

3 The difference Galois group

Lemma 4 (irreducibility of the σ -module). *For $0 < q < 1$ the rank-2 σ -module of G is irreducible over $\mathbb{C}(z)$: the Riccati equation*

$$u(z)u(Qz) - a(z)u(z) + q = 0, \quad a = q + 1 - qz,$$

has no solution $u \in \mathbb{C}(z)$.

Proof. Suppose $u = P/R$ with $P, R \in \mathbb{C}[z]$ coprime, of degrees p, r , leading coefficients α, β . Clearing denominators,

$$F(z) := P(Qz)P(z) - a(z)P(z)R(Qz) + qR(z)R(Qz) \equiv 0.$$

At $z = 0$: $F(0) = P(0)^2 - (q+1)P(0)R(0) + qR(0)^2 = 0$; if $R(0) = 0$ then $P(0) = 0$, contradicting coprimality, and symmetrically; so $P(0)R(0) \neq 0$ and $u(0) \in \{1, q\}$. *Top degree*: the three terms have degrees $2p, p+r+1, 2r$. If $p = r$, the top degree $2p+1$ is attained by the middle term alone, with coefficient $q\alpha\beta Q^r \neq 0$; if $|p-r| \geq 2$, the top degree is attained by the first or third term alone, with coefficient $\alpha^2 Q^p$ or $q\beta^2 Q^r$. Both are impossible, so $|p-r| = 1$. *Root pairing*: let $\zeta \neq 0$ be a root of R of multiplicity m , so $P(\zeta) \neq 0$. At $z = \zeta/Q$ the three terms of F have orders $k, k+m, m_0+m$, where $k = \text{ord}_{\zeta/Q} P$ and $m_0 = \text{ord}_{\zeta/Q} R$; since $F \equiv 0$ the minimum order cannot be attained by a single term, forcing $k \geq m_0 + m \geq m$. Symmetrically, a root $\xi \neq 0$ of P of multiplicity k forces $\text{ord}_{Q\xi} R \geq k$ (coprimality gives $\text{ord}_{Q\xi} P = 0$, so the first term has order k at ξ , the third $\text{ord}_{Q\xi} R$, and the middle at least $k + \text{ord}_{Q\xi} R$; again no single minimum). The two maps $\zeta \mapsto \zeta/Q, \xi \mapsto Q\xi$ are mutually inverse on the nonzero root sets, and chains close at length two by coprimality. Summing multiplicities, $p \leq r \leq p$, so $p = r$ — contradicting $|p-r| = 1$. $\square$

Theorem 5 (the difference Galois group contains SL_2). *For every $0 < q < 1$, the difference Galois group over $\mathbb{C}(z)$ of the module of G contains SL_2 .*

Proof. For $N \geq 1$ suppose $\text{Sym}^N M$ has a rank-1 submodule, i.e. there is a nonzero solution w of the N -th symmetric-power equation with $r := w(Q\cdot)/w \in \mathbb{C}(z)$. The products $f_1^{N-k} f_2^k$ ($0 \leq k \leq N$) form a solution basis (their Casoratian is a power of $\text{Cas}(f_1, f_2) \neq 0$; checked numerically for $N = 2$), so $w = \sum_k c_k f_1^{N-k} f_2^k$ with constants c_k ; set $k_0 := \min\{k : c_k \neq 0\}$.

Case $k_0 < N$. Along a ray in a regular class, f_2/f_1 decays super-polynomially, so the rational function r has the Poincaré asymptotic expansion of the quotient of the dominant component:

$$r \sim (-1/z)^{N-k_0} (-z/q)^{k_0} \left(\frac{\hat{g}(Qz)}{\hat{g}(z)} \right)^{N-k_0} \left(\frac{h(Qz)}{h(z)} \right)^{k_0}.$$

The j -th coefficient of the formal series on the right is $(N-k_0)b_j Q^{-j}(1 + O(q^{cj}))$ with $b_j = q^{j-j(j-1)/2}/(q;q)_j$: the dominant term comes from $b_j Q^{-j}$ in $\hat{g}(Qz)/\hat{g}(z)$, all competing products being smaller by a factor q^{cj} (verified: the ratio of the j -th coefficient to $(N-k_0)b_j(Q^{-j}-1)$ is $1 - 10^{-6}$ at $j = 12, q = \frac{1}{3}$). These coefficients grow like $q^{-j^2/2}$, whereas a rational function's expansion at ∞ has geometrically bounded coefficients. Since matching asymptotic expansions are unique, no rational r exists.

Case $k_0 = N$. Then $r = (-z/q)^N \rho^N$ with $\rho := h(Q\cdot)/h$ meromorphic on $\mathbb{C}^*$; $\rho^N \in \mathbb{C}(z)$ and single-valuedness force $\rho \in \mathbb{C}(z)$ (an N -th root of a rational function is single-valued on $\mathbb{C}^*$ only if every divisor multiplicity, including at 0 and ∞ , is divisible by N). Then $f_2(Qz)/f_2(z) = -(z/q)\rho(z)$ is rational, giving a rational solution of the rank-2 Riccati equation — contradicting Lemma 4.

Hence no $\text{Sym}^N M$ has a rank-1 submodule. $N = 1$ is irreducibility; $N = 2$ excludes the imprimitive (dihedral) case; and a projectively finite irreducible subgroup (A_4, S_4, A_5) fixes a line in

some Sym^N with $N \leq 12$ (the classical binary polyhedral invariants), which is likewise excluded. By the classification of algebraic subgroups of GL_2 , the group contains SL_2 . $\square$

Proposition 6 (no $\sigma\delta$ -integrability). *For every $0 < q < 1$ there is no $B \in \text{gl}_2(\mathbb{C}(z))$ and no $c \in \mathbb{C}(z)$ with $\theta(A) = \sigma(B)A - AB + cA$, where $A = \begin{pmatrix} 0 & 1 \\ -q & a \end{pmatrix}$ and $\theta = z\partial_z$.*

Proof. Writing $B = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$, the entries give $\gamma = -q\beta_1$, $\delta = \alpha_1 + a\beta_1 - c$, and eliminating α (the central terms c cancel identically; verified symbolically) leaves the necessary condition

$$qa\beta_3 = a_1(aa_1 - q)\beta_2 - a(aa_1 - q)\beta_1 + qa_1\beta + qz(a_1 + q^2a),$$

$\beta_k = \beta(Q^k z)$, $a = a(z)$, $a_1 = a(Qz)$. A polynomial part of degree $m \geq 0$ contributes top coefficient $\propto \beta_m(1 - q^{2+2m}) \neq 0$ at degree $m + 3$, above the inhomogeneity: so β has no polynomial part. If β has a pole off the orbit $z_a Q^{\mathbb{Z}}$, the maximal-modulus pole appears in exactly one term (the slot $z = \zeta Q^{-3}$, coefficient $qa \neq 0$ there), and the minimal-modulus pole likewise (slot $z = \zeta'$, coefficient qa_1): both impossible; so the poles lie in the window $\{z_a Q^{-1}, \dots, z_a Q^3\}$ ($a(z_a Q^k) = (q+1)(1 - Q^k)$ vanishes only at $k = 0$). Sweeping the slots $z = z_a Q^{-4}, \dots, z_a Q^{-1}$ kills the orders at $z_a Q^{-1}, \dots, z_a Q^2$ one by one (each pole appears in a single term with nonvanishing coefficient), and the order at $z_a Q^3$ is killed at the slot $z = z_a Q$ — coefficient $a_1(aa_1 - q)$ with $(aa_1 - q)(z_a Q) = (q+1)^2(1 - q^2)(1 - q^4) - q$ — or, where that vanishes (one root $q \approx 0.78027$), at the slot $z = z_a Q^2$ — coefficient $a(aa_1 - q)$ with $(aa_1 - q)(z_a Q^2) = (q+1)^2(1 - q^4)(1 - q^6) - q$ (one root $q \approx 0.86753$); the two polynomials are coprime, so for every $q \in (0, 1)$ at least one slot applies. Hence $\beta \equiv 0$, contradicting $z(a_1 + q^2a) = z[(q+1)(1 + q^2) - 2q^3z] \neq 0$. (Cross-check: at $q = \frac{1}{3}$ the exact linear system for poles of order ≤ 3 in the window has $\text{rank}(A) = 15$, $\text{rank}[A|b] = 16$.) $\square$

Corollary 7 (differential transcendence of G). *For every $0 < q < 1$ the function $G(q, \cdot)$ satisfies no algebraic differential equation over $\mathbb{C}(z)$: the Hahn–Exton q -cosine is differentially transcendental.*

Proof, via parametrized difference-Galois theory. We apply the parametrized (σ, θ) -difference Galois theory with derivation $\theta = z\partial_z$. The determinant equation $\sigma(y) = \det(A)y = qy$ has the solution $y = z^{1/2}$ (indeed $\sigma(z^{1/2}) = (Qz)^{1/2} = qz^{1/2}$, as $Q = q^2$), which is algebraic over $\mathbb{C}(z)$; hence the parametrized difference Galois group of $\sigma(y) = \det(A)y$ lies in $\mathbb{C}^\times$. This is exactly the hypothesis of Dreyfus–Hardouin–Roques [4, Prop. 2.9], which for a rank-2 system whose difference Galois group contains SL_2 (here Theorem 5) gives the dichotomy: the parametrized group Gal^θ either is conjugate into $\text{GL}_2(\mathbb{C})$ (all solutions differentially algebraic) or contains $\text{SL}_2(\tilde{\mathbb{C}})$ (all nonzero solutions differentially transcendental), and the *first* alternative holds if and only if there exists $B \in \text{gl}_2(\mathbb{C}(z))$ with $\theta(A) = \sigma(B)A - AB$ (their integrability condition (7); $\theta(\det A) = 0$ since $\det A = q$ is constant in z , so no trace term appears). By Proposition 6 no such B exists — indeed none exists even allowing a scalar twist cA . Therefore the second alternative holds: $\text{Gal}^\theta \supseteq \text{SL}_2(\tilde{\mathbb{C}})$, and every nonzero solution, in particular $G(q, \cdot)$, is differentially transcendental over $\mathbb{C}(z)$. $\square$

4 The connection constant and the zero product

The behaviour of G at $z = \infty$ is governed by a theta-comparison. Let $\theta_Q(x) = \sum_{n \in \mathbb{Z}} Q^{n(n-1)/2} x^n$, so $\theta_Q(Qx) = x^{-1}\theta_Q(x)$ and, by the Jacobi triple product, $\theta_Q(-z) = (Q; Q)_\infty(z; Q)_\infty(Q/z; Q)_\infty$, an entire function of order 0 with simple zeros exactly at $z \in Q^{\mathbb{Z}}$. The following theorem evaluates the leading asymptotic constant of G and, as a consequence, the regularized product of its zeros; the constant $C_1 := \lim_{R \rightarrow \infty} G(-R)/\theta_Q(R)$ coincides with the Stokes-invariant connection coefficient of G against the theta solution.

Theorem 8 (the connection constant, and the zero product of the Hahn–Exton q -cosine). For $0 < q < 1$,

$$C_1 = \frac{1}{(q; q)_\infty}, \quad \text{and consequently} \quad \prod_{k \geq 1} z_k(q) Q^{k-1} = (q; q^2)_\infty,$$

where $z_1 < z_2 < \dots$ are the positive zeros of $G(q, \cdot)$. Likewise, for the second solution's underlying entire function H with zeros $w_1 < w_2 < \dots$, $\prod_{k \geq 1} w_k(q) q Q^{k-1} = (q^3; q^2)_\infty$. (Verified at three bases including q^* , to 41, 23, 26 digits for C_1 and 58 digits for the product.)

Proof. (1) *An exact identity.* From $(q; q)_{2k} = (q; q)_\infty / (q^{2k+1}; q)_\infty$ and Euler's expansion $(x; q)_\infty = \sum_{j \geq 0} (-1)^j q^{j(j-1)/2} x^j / (q; q)_j$ at $x = q^{2k+1}$, together with $k(k-1) + 2kj = (k+j)(k+j-1) - j(j-1)$ and the substitution $m = k + j$, absolute convergence of the double sum permits the rearrangement

$$(q; q)_\infty G(z) = \sum_{m \geq 0} (-1)^m q^{m(m-1)/2} z^m P_m(1/z), \quad P_m(w) := \sum_{j=0}^m c_j w^j, \quad c_j = \frac{q^{j-j(j-1)/2}}{(q; q)_j}$$

(the c_j coincide with the b_j of §2; identity checked symbolically through z^8 and numerically to 41 digits).

(2) *The constant.* Fix $u < 0$ and put $z = uQ^{-N}$. The moduli $q^{m(m-1)/2}|z|^m$ peak at $m_0 = N + O(1)$ and decay like $q^{(m-m_0)^2 + O(1)}$ away from it. For $1 \leq j \leq m \leq 2N$, $2Nj - j(j-1)/2 \geq Nj$, so $|c_j z^{-j}| \leq Cq^{Nj}|u|^{-j}$ and $|P_m(1/z) - 1| \leq C'q^N$ uniformly; the terms with $m > 2N$ contribute a relative $O(q^{N^2})$. The negative- m part of $\theta_Q(-z)$ is $O(|z|^{-1})$, negligible against the peak. Hence $(q; q)_\infty G(z) / \theta_Q(-z) \rightarrow 1$ along $z = uQ^{-N}$, for each fixed $u < 0$.

(3) *Geometric zero dressing.* On the circle $|z - Q^{1-k}| = \varepsilon q^k Q^{1-k}$ the identity of step (1) and the bounds of step (2) give $|(q; q)_\infty G - \theta_Q(-z)| \leq Cq^k |\theta_Q(-z)|_{\text{peak}}$, while $\theta_Q(-z)$ has a simple zero at Q^{1-k} with derivative bounded below there; Rouché gives $z_k = Q^{1-k}(1 + O(q^k))$, i.e. $\log(z_k Q^{k-1}) = O(q^k)$, absolutely summable.

(4) *The product.* $G(z) = \prod_k (1 - z/z_k)$ (order 0, genus 0, $G(0) = 1$) and $\theta_Q(-z) = (Q; Q)_\infty (z; Q)_\infty (Q/z; Q)_\infty$ (Jacobi). Along $z = -R$, $R \rightarrow \infty$,

$$\frac{G(-R)}{\theta_Q(R)} = \frac{1}{(Q; Q)_\infty (-Q/R; Q)_\infty} \prod_{k \geq 1} \frac{1 + R/z_k}{1 + RQ^{k-1}},$$

and for $x, y > 0$ the function $R \mapsto \log \frac{1+Rx}{1+Ry}$ is monotone from 0 to $\log(x/y)$, so each factor is dominated by $|\log(z_k Q^{k-1})|$, summable by step (3). Dominated convergence gives $G(-R)/\theta_Q(R) \rightarrow 1/[(Q; Q)_\infty \prod_k z_k Q^{k-1}]$ as $R \rightarrow \infty$; evaluating along the subsequence $R = |u|Q^{-N}$ by step (2), the limit equals $1/(q; q)_\infty$, whence $C_1 = 1/(q; q)_\infty$ and $\prod_k z_k Q^{k-1} = (q; q)_\infty / (Q; Q)_\infty = (q; q^2)_\infty$ by Euler's $(q; q)_\infty = (q; q^2)_\infty (q^2; q^2)_\infty$. The H -statement is identical with comparison function $\theta_Q(-qz)$ and ratio $(1 - q)/(q; q)_\infty$. $\square$

Remark 9 (the modular split at z^*). At $q = q^*$ the product splits off the candidate rational zero:

$$\prod_{k \geq 2} z_k(q^*) Q^{k-1} = \frac{(q; q^2)_\infty}{z^*} = 0.988299884628746\dots \quad (\text{verified to 30 digits}).$$

Were z^* (equivalently β_2) algebraic, the modular value $(q; q^2)_\infty$ — transcendental at algebraic q by Nesterenko's theorem [7] applied to the Euler products — would equal an algebraic number times the tail product $\prod_{k \geq 2} z_k Q^{k-1}$. This is consistent only if the tail product is itself transcendental; the arithmetic of the zero condition is thereby confined to that tail, which is the content of the multiplicity problem of [3].

5 The arithmetic of the zeros

5.1 Conjecture Z and its confluent evidence

Conjecture 3 is the q -analogue of the classical theorem (Siegel; see also Lindemann for the endpoint) that the positive zeros $j_{\nu,k}$ of the Bessel function J_ν are transcendental for rational ν . Two structural facts support it. First, Theorem 1: G is differentially transcendental, so its zeros are not the zeros of any algebraic-differential shadow whose arithmetic could be controlled classically. Second, in the confluent limit $q \rightarrow 1$ the equation (1) degenerates to Bessel's equation and $z_k(q) \rightarrow (qj_{-1/2,k})^2$, so Conjecture 3 deforms Siegel's theorem.

That confluence also explains, in one line, why the classical case is provable and the q -case is not. Write D_N for the denominator of the N -th partial sum. The classical engine (Lambert, Lindemann, and every factorial-denominator proof) runs on a *divergent step ratio*: $D_{N+1}/D_N = N+1 \rightarrow \infty$, so one further term is gained at asymptotically no arithmetic cost and $\text{tail} \cdot D_N \asymp 2/(N+1) \rightarrow 0$ (measured: 0.310, 0.163, 0.099 at $N = 3, 6, 10$ for $e = \sum 1/k!$). The q -deformation replaces the factorial by $(q; q)_k$, whose step ratio is *constant* in the relevant scale: each further term gains $\asymp q^{2N}$ but costs $\asymp t^{2N}$, and at $s = 1$ the two cancel identically, so $\text{tail} \cdot D_N$ diverges (measured at $q = \frac{1}{3}$: 7.1, $7.5 \cdot 10^5$, $5.1 \cdot 10^{14}$ at $N = 2, 4, 6$). The bridge is explicit: $(q; q)_k / ((1-q)^k k!) \rightarrow 1$ as $q \rightarrow 1$ (measured 0.472, 0.928, 0.993 at $q = 0.9, 0.99, 0.999$). The factorial — and with it the divergent step ratio that powers Lindemann — exists only in the confluent limit; at any $q < 1$ the ratio is constant, and a constant ratio can never win. This is the same exactness of Remark 19 seen dynamically, and it identifies precisely what a proof of Conjecture 3 must supply: an arithmetic mechanism with no classical shadow.

5.2 The self-referential specialization: β_2

The reflection-group growth constant β_2 is the reciprocal square root of the least positive root q^* of $S(q) = G(q, 2q(1-q))$, which is exactly the condition that the catalytic point $z^* = 2q^*(1-q^*)$ be a zero of $G(q^*, \cdot)$. Thus β_2 is governed by a *self-referential* specialization of Conjecture 3: the zero and the base move together along the diagonal $z = 2q(1-q)$.

At this point the q -Wronskian (2) yields a clean exact relation between two Hahn–Exton values: since $G(q^*, z^*) = 0$,

$$G(q^*, q^{*2}z^*) \cdot H(q^*, z^*) = 1 - q^* \quad (3)$$

(verified to 25 digits) — a product of two q -Bessel values at the algebraic point z^* equal to the algebraic number $1 - q^*$. Together with $G(q^*, z^*) = 0$ and the modular zero product of Theorem 2, this places three exact constraints on the values at z^* ; a transcendence proof requires one more independent constraint than the reductio supplies, which is the precise sense in which Conjecture 3 is a multiplicity (rather than a single-relation) problem.

5.3 The q -Siegel–Shidlovskii margin

The natural route to Conjecture 3 is the q -analogue of the Siegel–Shidlovskii method developed by Amou–Matala-aho–Väänänen [1]. That method applies to Poincaré-type q -difference systems — a hypothesis (1) satisfies at $z = 0$ — under a *Siegel margin* $\gamma < \Gamma$, where γ is the ratio of the logarithmic height of the cleared denominators to that of the analytic decay. For the Hahn–Exton class this ratio is

$$\gamma = \frac{\log(t/s)}{2 \log t} \leq \frac{1}{2} \quad (q = s/t \text{ in lowest terms}),$$

the factor 2 being the double Pochhammer $(q; q)_{2k}$; this is below the threshold, so the method does not apply as stated. Extending it past the margin for this irregular-at- ∞ class is the outstanding problem; the connection data needed for the irregular direction is available in closed form [3].

5.4 Integrality of the zero lattice

The zeros, viewed as functions of the base, are integer power series after the Newton-polygon rescale — all of them, not only the first.

Theorem 10 (integrality of the zero lattice). *For every $k \geq 1$ the rescaled k -th zero $u_k(q) := q^{2(k-1)} z_k(q)$ lies in $1 + q\mathbb{Z}[[q]]$. In particular*

$$z_1(q) = 1 - q - q^3 + q^4 - q^5 + 2q^6 - 2q^7 + 4q^8 - 6q^9 + 8q^{10} - 14q^{11} + 21q^{12} - \dots \in \mathbb{Z}[[q]].$$

Proof. From $g_{k'} = (-1)^{k'} q^{k'(k'-1)} / (q; q)_{2k'}$ and the exponent identity $k'(k'-1) - 2(k-1)k' + k(k-1) = (k' - k)(k' - k + 1)$,

$$F_k(q, u) := q^{k(k-1)} G(q, u q^{-2(k-1)}) = \sum_{k' \geq 0} (-1)^{k'} q^{(k'-k)(k'-k+1)} \frac{u^{k'}}{(q; q)_{2k'}}.$$

The exponent $(k' - k)(k' - k + 1)$ is a product of consecutive integers, hence nonnegative, and vanishes exactly at $k' \in \{k-1, k\}$: each zero scale is carried by its own edge of the strictly convex Newton polygon of G . Since $1/(q; q)_{2k'} \in \mathbb{Z}[[q]]$ (partitions into parts $\leq 2k'$), $F_k \in \mathbb{Z}[[q]][[u]]$ with $F_k(0, u) = (-1)^{k-1} u^{k-1} (1 - u)$; at $u = 1$ the value vanishes and $\partial_u F_k(0, 1) = (-1)^k$ is a unit, so Hensel's lemma in the complete local ring $\mathbb{Z}[[q]]$ yields a unique root $u_k \in 1 + q\mathbb{Z}[[q]]$, which by the scale localization $z_k = q^{2(1-k)}(1 + O(q^k))$ of §4 is the k -th zero. (Verified to order q^{300} for $k \leq 4$: $u_2 = 1 - q^6 - 2q^8 + q^9 - \dots$, $u_3 = 1 - q^{15} - \dots$, $u_4 = 1 - q^{28} - \dots$. The onset law visible here is proved next.) $\square$

5.5 Lattice values and the hexagonal law

The deviation $u_k - 1$ begins exactly at the k -th hexagonal number $k(2k-1)$. This reduces to, and is proved by, an exact evaluation of G at its own lattice points.

Lemma 11 (parameter–argument swap). *For $|Q| < 1$, $(b; Q)_\infty {}_1\phi_1(0; b; Q, z)$ is symmetric in (b, z) ; equivalently*

$$(q; Q)_\infty G(q, z) = \sum_{n \geq 0} \frac{(-1)^n q^{n^2}}{(Q; Q)_n} (zQ^n; Q)_\infty. \quad (4)$$

Proof. In $(b; Q)_\infty {}_1\phi_1(0; b; Q, z) = \sum_n (-1)^n Q^{\binom{n}{2}} z^n (bQ^n; Q)_\infty / (Q; Q)_n$ expand $(bQ^n; Q)_\infty = \sum_i (-1)^i Q^{\binom{i}{2}} (bQ^n)^i / (Q; Q)_i$ (Euler); since $\binom{n}{2} + \binom{i}{2} + ni = \binom{n+i}{2}$, the double sum is $\sum_{n,i} (-1)^{n+i} Q^{\binom{n+i}{2}} z^n b^i / ((Q; Q)_n (Q; Q)_i)$, manifestly symmetric. For (4) take $b = q$ and use $(z; Q)_\infty / (z; Q)_n = (zQ^n; Q)_\infty$ on the swapped side, with $Q^{\binom{n}{2}} q^n = q^{n^2}$. (Verified to 40 digits at generic (b, z) .) $\square$

Proposition 12 (square vanishing at the lattice). *For every $m \geq 0$,*

$$G(q, Q^{-m}) = (-1)^{m+1} \frac{(Q; Q)_\infty}{(q; Q)_\infty} \sum_{r \geq 0} \frac{(-1)^r q^{(m+1+r)^2}}{(Q; Q)_r (Q; Q)_{m+1+r}},$$

so $\text{ord}_q G(q, Q^{-m}) = (m+1)^2$ exactly, with leading coefficient $(-1)^{m+1}$.

Proof. Evaluate (4) at $z = Q^{-m}$. The factor $(Q^{n-m}; Q)_\infty$ vanishes for $n \leq m$ and equals $(Q; Q)_\infty / (Q; Q)_{n-m-1}$ for $n \geq m+1$; reindexing $n = m+1+r$ gives the display. The q -order of the r -th term is $(m+1+r)^2$, strictly increasing in r , and all other factors are units in $\mathbb{Z}[[q]]$, so no cancellation can occur and the $r=0$ term decides. (Verified as an identity in $\mathbb{Z}[[q]]$ to order 250 for $m \leq 4$, and to 50 digits at $q = 0.3$.) $\square$

Corollary 13 (hexagonal law). *For every $k \geq 1$,*

$$u_k(q) = 1 - q^{k(2k-1)}(1 + O(q)) :$$

the deviation of the k -th rescaled zero from 1 begins exactly at the k -th hexagonal number, with leading coefficient -1 .

Proof. $F_k(q, 1) = q^{k(k-1)}G(q, Q^{1-k}) = (-1)^k q^{k(2k-1)}(1 + O(q))$ by Proposition 12 at $m = k-1$ (note $k(k-1) + k^2 = k(2k-1)$), while $\partial_u F_k(q, 1) = (-1)^k(1 + O(q))$. The first Newton correction from $u = 1$ is therefore $-q^{k(2k-1)}(1 + O(q))$, and all subsequent corrections are $O(q^{2k(2k-1)})$. This proves the onsets 1, 6, 15, 28 observed above and predicts $u_5 = 1 - q^{45} - \dots$, confirmed directly. $\square$

Remark 14. Three readings of Proposition 12. (a) The r -sum is again a Hahn–Exton series, now of integer order $m+1$ in base Q : the q -cosine ($\nu = -\frac{1}{2}$) sampled on its own zero lattice reproduces the integer-order family — an order reflection $-\frac{1}{2} \mapsto m+1$. (b) Along any backward orbit $z = Q^{-m}w$, (4) gives $G(Q^{-m}w) \sim (Q^{-m}w; Q)_\infty / (q; Q)_\infty$ as $m \rightarrow \infty$, with constant $(w; Q)_\infty (Q/w; Q)_\infty / (q; Q)_\infty = \theta_Q(-w)/(q; q)_\infty$ by the triple product — recovering the connection constant of the backward normal form independently. (c) At any zero $z_k(q)$ of G , (4) transfers the vanishing into the *parameter* slot: ${}_1\phi_1(0; z_k(q); Q, q) = 0$. At the crossing $z_1(q^*) = 2q^*(1 - q^*)$ of the reductio this is a ${}_1\phi_1$ -vanishing with all-rational parameters; its truncation ledger has the same heights as the original (the swap preserves the exactness of Remark 19).

Remark 15 (arithmetic and analytic features of the zero curve). (a) The series is not rational and satisfies no algebraic relation $P(q, z_1) = 0$ of bidegree $\leq (6, 5)$ (exact linear algebra on 57 coefficients). (b) Its radius of convergence is $|q_c| = 0.5545\dots$, set by the real fold point of item (e); in particular $q^* = 0.4494\dots$ lies inside the disc of convergence. (c) The integrality opens a *linear-height* approximation ledger: at $q = s/t$ the N -th partial sum is rational with denominator t^N and error $\asymp (q/R)^N$, $R = 0.5545\dots$, so the Liouville criterion for the crossing $z_1(q) = 2q(1 - q)$ reads $s < R$ — failing, but now only by the factor $s/R \approx 1.80$ at $s = 1$, the narrowest margin among all ledgers recorded for this problem; the obstruction has become the sub-unit radius, i.e. the complex collision of zeros. (d) The integer coefficient sequence 1, -1 , 0, -1 , 1, -1 , 2, -2 , 4, -6 , 8, -14 , 21, -34 , 56, -88 , 148, -242 , 398, -669 , 1109, $\dots$ appears to be new.

(e) The radius-limiting singularity is located and characterized: it is the simple fold $q_c = -0.5545786101465797\dots$, $z_c = 2.5239525204417103\dots$ (both to 230 digits in the repository), where the analytically continued branch collides with a second zero branch on the negative q -axis ($G = G_z = 0$ with $G_{zz} = 0.2143\dots \neq 0$, $G_q = -1.5012\dots \neq 0$, so $z_1(q) = z_c \pm \tilde{C}\sqrt{q - q_c} + \dots$ with $\tilde{C}^2 = 2G_q/(-G_{zz})$ at the fold). The fold law now has a closed-form constant:

$$|c_n| n^{3/2} |q_c|^n \longrightarrow \frac{|\tilde{C}| \sqrt{|q_c|}}{2\sqrt{\pi}} = 0.7863083135\dots,$$

confirming $R = |q_c|$ exactly (the sequence reaches 0.78295 at $n = 299$ with a clean $1/n$ tail extrapolating to 0.78621; an earlier reading 0.769 from $n \leq 120$ was preasymptotic). Neither q_c nor z_c satisfies an integer polynomial relation of degree ≤ 8 with coefficients up to 10^{14} , nor does the

pair satisfy a bilinear one (PSLQ at 230 digits, tolerance 10^{-200} ; degree-8 candidates returned at 120 digits were exposed as spurious by re-evaluation at 230).

(f) No change of variable rescues the ledger of (c): if $q = \psi(u)$ with $\psi \in u\mathbb{Z}[[u]]$ (as required to keep $z_1 \circ \psi \in \mathbb{Z}[[u]]$) and $u^* = \varphi(q^*) = s'/t' \in \mathbb{Q}$, the criterion again reads $|u^*|t' = s' < R_u$; but a non-polynomial integer power series has radius at most 1, so $R_u \leq 1$ while $s' \geq 1$. Every integral uniformization is capped at the boundary case $s' = 1 = R_u$: the sub-unit radius can be moved but not beaten. This is the geometric face of the same boundary on which the theta and Rogers–Ramanujan series sit in the truncation ledger (Remark 19).

(g) The cap has an elementary final form, the *discreteness cap*: for any non-polynomial $X \in \mathbb{Z}[[q]]$, any $q_0 = s/t$ inside its disc of convergence (away from the boundary), and any N , the truncation tail satisfies $|X(q_0) - X_N(q_0)| = |c_M|(s/t)^M(1 + o(1)) \geq s^M t^{-M}(1 - o(1))$, where $M > N$ is the first index with $c_M \neq 0$ — at least the Liouville floor t^{-M} of the truncation’s denominator whenever $s \geq 1$, because a nonzero integer coefficient has $|c_M| \geq 1$. (Measured on z_1 at $q = \frac{1}{7}$: tail/floor = 1.53, 11.6, 70.9, 541 at $N = 6, 10, 14, 18$.) Integrality thus caps its own ledger: the property that grants linear heights also forbids the tail from undercutting the floor. (h) The arithmetic class of z_1 is now pinned from above as well: it is *not holonomic*. No linear differential equation $\sum_{i \leq r} P_i(q)z_1^{(i)} = 0$ with $\deg P_i \leq d$ exists for (r, d) up to $(8, 10)$, $(6, 14)$, $(5, 12)$ (exact linear algebra on 110–116 of the integer coefficients, computed to order 120; every coefficient matrix has full column rank). Thus z_1 is an integer power series that is simultaneously non-algebraic, non-holonomic, and of radius < 1 — precisely the class for which no arithmetic technology exists: the Chudnovsky–André–Bost theory and the recent arithmetic holonomy bounds of Calegari–Dimitrov–Tang, which are built for “integrality plus structure”, all require holonomy, while Pólya–Carlson requires radius exactly 1. Integrality alone, without holonomy, is not currently actionable.

(i) One exact relation ties the lattice across the bases q , $-q$, q^2 : applying the zero product of Theorem 2 at the three bases and using $(q; q^2)_\infty(-q; q^2)_\infty = (q^2; q^4)_\infty$,

$$\prod_{k \geq 1} \varphi_k = 1, \quad \varphi_k := \frac{u_k(q) u_k(-q)}{u_k(q^2)}.$$

The relation is not termwise: $\varphi_1 = 1 + 2q^6 + 4q^8 + 8q^{10} + 18q^{12} + \dots$, $\varphi_2 = 1 - 2q^6 - \dots$, $\varphi_3 = 1 + 2q^{20} + \dots$, $\varphi_4 = 1 - 2q^{28} - \dots$ (order 300; adjacent leading defects cancel in pairs). No Mahler structure underlies it: u_1 satisfies no polynomial relation $P(q, u_1(q), u_1(q^d)) = 0$ for $d = 2, 3$ with degrees up to $(24, 2, 2)$, $(16, 3, 3)$, $(10, 4, 4)$; the system $\{u_1, u_2, u_3\}$ admits no relation of the analogous bilinear profile (exact linear algebra on the 300 integer coefficients, all matrices of full column rank); and no zero of the base- q^2 cosine or sine is a rational multiple of $q^e z_i(q)^a w_j(q)^b$ (or the sine-side analogue) for $|e| \leq 10$, $a + b \leq 2$ (numerical search at 60 digits, cross-checked at two bases). Mahler’s method — intrinsically q -analytic, with no classical shadow — thus finds no foothold on the zero curve at these degrees.

(j) The fold is the natural point for a rationality reductio, because it is the unique point of the (q, z) -plane carrying *two* vanishings ($G = \theta G = 0$; a triple point $G = \theta G = \theta^2 G = 0$ is codimension three in two variables, hence absent). The first-jet web there consists of six values, $A = G(Qz_c)$, $B = \theta G(Qz_c)$, $h = H(z_c)$, $h_1 = H(Qz_c)$, $\eta = \theta H(z_c)$, $\eta_1 = \theta H(Qz_c)$, where H is the sine-side companion, which satisfies the mirror equation $H(z) - a(z)H(Qz) + qH(Q^2z) = 0$ (the $z^{1/2}$ in the second solution $z^{1/2}H$ moves the outer q). Exactly two finite-height relations hold: the Casoratian at z_c , $Ah = 1 - q_c$, and its θ -derivative, $Bh = -A\eta$ (both verified to 236 digits). For any mixed jet-Casoratian $C_{u,v}(z) = qu(z)v(Qz) - u(Qz)v(z)$ with $L(u) = f$, $\tilde{L}(v) = \tilde{g}$ the a -terms cancel in the increment:

$$C_{u,v}(Qz) - C_{u,v}(z) = \tilde{g}(z)u(Qz) - f(z)v(Qz),$$

so every telescoped sum of first-jet orbit bilinears closes back onto a Casoratian. In particular $B\eta = \sum_{m \geq 0} qQ^m z_c [G\theta H - H\theta G](Q^{m+1} z_c)$ (verified to 121 digits) is exact but carries an infinite rational series as coefficient — height-infinite. The ledger of a point evaluation is therefore fixed: each further θ - or δ -jet level adds at least three new values and exactly one finite-height relation, so the deficit (four at the fold, five at a generic zero) cannot be closed by jet-raising, and the fold's two vanishings are the maximum the plane admits. Point evaluation alone, at any point including the fold, cannot complete a rationality reductio for this module.

Together, (c), (f), (g) and Remark 19 express one law from four sides: for every representation of this problem yet constructed, the arithmetic cost equals the analytic decay *exactly*, with zero excess; a proof of Conjecture 3 must produce strict excess, and rearrangement, reparametrization, or symmetrization of the object's own $\mathbb{Q}(q)$ -structure provably cannot.

5.6 The second-kind forms and the exact reduction

The q -Siegel construction for this class is explicit. For $N \geq 1$ put

$$P_N(q, z) := (q; q)_{2N} \sum_{k=0}^N \frac{(-1)^k q^{k(k-1)} z^k}{(q; q)_{2k}} = \sum_{k=0}^N (-1)^k q^{k(k-1)} z^k \prod_{i=2k+1}^{2N} (1 - q^i) \in \mathbb{Z}[q, z].$$

Proposition 16 (the second-kind forms). $\deg_q P_N = 2N^2 + N$ and $\deg_z P_N = N$ (exact; verified $N \leq 4$), and the remainder $R_N(q, z) := (q; q)_{2N} G(q, z) - P_N(q, z)$ satisfies, uniformly for q in compacts of $(0, 1)$ and z bounded,

$$|R_N(q, z)| = q^{N^2(1+o(1))}.$$

Proof. Membership in $\mathbb{Z}[q, z]$ and $\deg_z P_N = N$ are clear from the displayed product form, since $(q; q)_{2N}/(q; q)_{2k} = \prod_{i=2k+1}^{2N} (1 - q^i) \in \mathbb{Z}[q]$. For the q -degree, the k -th summand has degree $k(k-1) + \sum_{i=2k+1}^{2N} i = 2N^2 + N - k^2 - 2k$, maximal at $k = 0$ with value $\deg(q; q)_{2N} = 2N^2 + N$; the maximum is attained once, so no cancellation occurs. For the remainder, $R_N = (q; q)_{2N} \sum_{k > N} (-1)^k q^{k(k-1)} z^k / (q; q)_{2k}$, whose leading term ($k = N+1$) is $\pm q^{N(N+1)} z^{N+1} (q; q)_{2N} / (q; q)_{2N+2}$; the tail is dominated by a geometric series in q^{2N} , so $|R_N| = q^{N^2+O(N)}$ uniformly. (Measured $\log |R_N|/N^2 \rightarrow \log q$ at both $z = 2q(1-q)$ and a generic z_0 .) $\square$

Proposition 17 (reduction to the Siegel margin). *Let $q = s/t \in (0, 1)$ and let $z_0 = a/b \in \mathbb{Q}^\times$ with $G(q, z_0) = 0$. Then $P_N(q, z_0) = -R_N(q, z_0)$ and $t^{2N^2+N} b^N P_N(q, z_0) \in \mathbb{Z}$. Consequently, if for infinitely many N one has $P_N(q, z_0) \neq 0$ and*

$$|R_N(q, z_0)| < t^{-(2N^2+N)} b^{-N},$$

then $G(q, z_0) \neq 0$ — a contradiction. By Proposition 16 the displayed criterion reads $q^{N^2} < t^{-2N^2}$ up to $O(N)$ in the exponent, i.e.

$$\log(t/s) > 2 \log t \iff st < 1,$$

which is false. The deficit is exactly the factor 2 in $\deg_q P_N = \deg(q; q)_{2N}$: the double Pochhammer.

Proof. $P_N(q, z_0) = (q; q)_{2N} G(q, z_0) - R_N(q, z_0) = -R_N(q, z_0)$ by hypothesis; clearing denominators of $P_N \in \mathbb{Z}[q, z]$ at $q = s/t$, $z = a/b$ gives $t^{\deg_q P_N} b^{\deg_z P_N} P_N(q, z_0) \in \mathbb{Z}$, and a nonzero integer has modulus ≥ 1 . The stated evaluation of the criterion is Proposition 16. $\square$

Corollary 18 (what would suffice). *Conjecture 3 at z_0 — hence, on the diagonal, the irrationality of β_2 — follows from any family of forms $\Lambda_N \in \mathbb{Z}[q, z]$ with $\Lambda_N(q, z_0) \neq 0$, remainder $q^{N^2(1+o(1))}$ against G , and*

$$\deg_q \Lambda_N \leq N^2 \frac{\log(t/s)}{\log t} (1 - \varepsilon), \quad \text{in particular } \deg_q \Lambda_N \leq N^2(1 - \varepsilon) \text{ when } s = 1.$$

The forms of Proposition 16 have $\deg_q = 2N^2$: it suffices to halve the q -degree at the same remainder, or equivalently to double the remainder decay to q^{2N^2} . A single-Pochhammer clearing $(q; q)_N$ would have degree $\frac{1}{2}N^2(1+o(1))$ and would suffice at once; no such representation of G along a non- q -power argument exists in the Rogers–Ramanujan/Bailey framework.

Remark 19 (the theta-decay barrier: why Corollary 18 is unreachable by truncation). The hypothesis of Corollary 18 cannot be met by *any* truncate-and-clear form. Let $\sum_k c_k(q) q^{\alpha k^2} z^k$ ($\alpha > 0$, $c_k \in \mathbb{Q}(q)$) be truncated at N and cleared. The $k = N$ summand already contributes $q^{\alpha N^2}$, so $\deg_q \geq \alpha N^2$; the remainder is the first omitted term, of size $q^{\alpha N^2(1+o(1))}$. The criterion $|R_N| < t^{-\deg_q}$ therefore reads

$$\alpha N^2 \log(t/s) > \alpha N^2 \log t \iff s < 1,$$

false for every $q = s/t \in (0, 1)$: the decay and the degree are driven by the same factor $q^{\alpha k^2}$ and cancel exactly. Denominators only make matters worse. Numerically, the cleared truncations have $\deg_q = N^2$ exactly for the theta series $\sum q^{k^2} z^k$ and for the Rogers–Ramanujan series $\sum q^{k^2} / (q; q)_k$ (both $\beta := \deg_q / N^2 = 1$, the boundary), while the Hahn–Exton series sits at $\beta \rightarrow 2$ (2.25, 2.17, 2.12 at $N = 4, 6, 8$) — a further factor 2 past a boundary that is itself already unreachable.

This explains the shape of the known landscape rather than merely our position in it: the transcendence of theta values and of Rogers–Ramanujan values — both at $\beta = 1$ — is known *only* through Nesterenko’s modular method [7], never through a Siegel/Liouville margin, precisely because no theta-decay series can cross $\beta < 1$. Consequently the only mechanisms available for Conjecture 3 are modularity, which G provably lacks (it is off the modular class; see [3]), or a multiplicity theory for the associated (σ, δ) -module. The double Pochhammer is not the whole obstruction — it is a factor 2 on top of a structural one.

5.7 An unconditional effective bound

Although Conjecture 3 is open, the specific root q^* admits an effective non-rationality statement, proved by certified interval arithmetic.

Proposition 20 (effective non-rationality of q^*). *The least positive root q^* of S admits no rational representation s/t with $t \leq 10^{191}$. Consequently, if $\beta_2 = u/v$ in lowest terms then $u > 10^{95}$.*

Proof. All steps are certified interval arithmetic (outward rounding; `mpmath.iv`; script `beta2_effective_irrationality` with the series tail bounded by $(q; q)_{2k} \geq (1 - 2q)/(1 - q) > 0$ for $q < \frac{1}{2}$ and term ratios $< \frac{1}{2}$ beyond the truncation. (i) $S > 0$ on $[0, 0.40]$ (400 certified chunks). (ii) $S' \leq -1$ on $[0.40, 0.46]$ (600 certified chunks; S' evaluated term-by-term with the logarithmic-derivative weights), so S has at most one zero in $[0.40, 0.46]$. (iii) with $a = 0.d_1 \dots d_{385}$ the 385-digit truncation of q^* and $b = a + 2 \cdot 10^{-385}$, certified evaluation gives $S(a) > 0 > S(b)$; by (i)–(ii), $q^* \in [a, b]$. (iv) the interval $[a, b]$ contains the rational s_0/t_0 (Stern–Brocot minimal fraction) with $t_0 \geq 10^{192}$, verified by exact rational comparison, and certified $S(s_0/t_0) \neq 0$. If $m/n \in [a, b]$ with $n \leq 10^{191}$ and $m/n \neq s_0/t_0$, then $|m/n - s_0/t_0| \geq 1/(nt_0) \geq 10^{-384} > b - a$, impossible; so $m/n = s_0/t_0$, whence $n = t_0 > 10^{191}$, a contradiction. Thus no rational with denominator $\leq 10^{191}$ lies in $[a, b]$ at all, in particular none equals q^* . For $\beta_2 = u/v$ in lowest terms, $q^* = v^2/u^2$ in lowest terms, so $u^2 > 10^{191}$. $\square$

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